

Lecture 13

18/09/24

10:07 AM

• CONDITIONAL EXPECTATION $E(X|Y=y)$

$$E(X|Y=y) = \sum_x x \cdot p_{X|Y}(x|y)$$

$$E(X) = \sum_y p_Y(y) \cdot E(X|Y=y)$$

* Similarly, for conditioning continuous r.v.s,

$$\begin{aligned} p_X &\rightarrow f_X \\ \sum &\rightarrow \int \end{aligned}$$

refer slides

• CONDITIONAL EXPECTATION $E(X|Y)$

y is not taking a specific value

$\therefore E(X|Y)$ is a function of y

$$g(Y) = E(X|Y)$$

$$\begin{aligned} \text{e.g. } g(y_1) &= E(X|Y=y_1) \\ g(y_2) &= E(X|Y=y_2) \end{aligned}$$

$$\begin{aligned} E(X|Y) \\ \downarrow \text{fn of } y \end{aligned}$$

LAW OF ITERATED EXPECTATIONS

$$E(E(X|Y)) = \sum_{y_i} p_Y(y_i) \cdot E(X|Y=y_i) = E(X)$$

$$E(E(X|Y)) = E(X)$$

* iid = independent and identically distributed.

Law of iterated variances?

• Sums of independent random variables

$$Z = X + Y$$

$$p_Z(z) = \sum_{\{(x,y): x+y=z\}} p_{X,Y}(x,y)$$

$$f_Z(z) = \int_{\{(x,y): x+y=z\}} f_{X,Y}(x,y) dx dy$$

CONVOLUTION FORMULA:

$$p_Z(z) = \sum_x p_X(x) \cdot p_Y(z-x)$$

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) \cdot f_Y(z-x) dx$$